

Project Description: Comparative Analysis of Adiabatic Error Bounds in Quantum Systems Through a Modified Landau-Zener Model

1 MOTIVATION AND BACKGROUND

The adiabatic theorem describes the evolution of a quantum system whose degrees of freedom vary slowly in time [1, pp. 419-420]. In contrast, diabatic evolution induces unwanted transitions that drive population into non-target states and disturb the prepared state. This continuous-time evolution can be sufficiently discretized in time which approximate the real dynamics. The adiabatic theorem can be extended to quantum computing, where the system is prepared in the ground state of an initial Hamiltonian which evolves according to the time-dependent Schrödinger equation, with the time dependence entering through the Hamiltonian itself. This continuous-time Schrödinger evolution can later be shifted to a discretized regime via a sequence of unitary (walk) operators. This discretized adiabatic formulation is then expressed directly in terms of these unitaries, without requiring an explicit Hamiltonian description, making it suited to stepwise implementations.

A key requirement is that the spectral gap between the instantaneous ground state energy and the rest of the spectrum (in particular, the first excited level) remains non-zero throughout the evolution. When this holds and the interpolation is sufficiently slow, transitions out of the ground state are suppressed and the system closely follows the instantaneous ground state at all times. This regime is referred to as adiabatic evolution, in which the system remains adiabatic throughout the process.

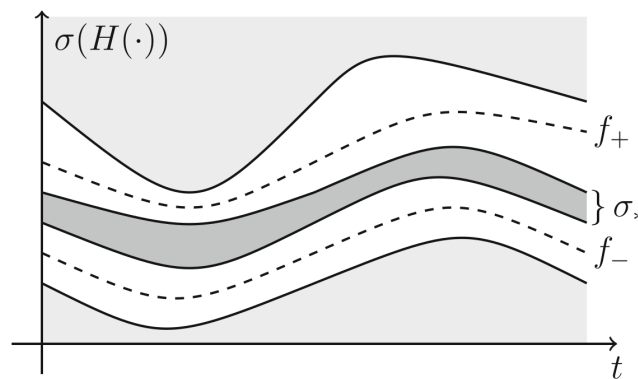


Figure 1: Sketch of the energy spectrum. An isolated subset of the spectrum $\sigma_*(t) \subset \sigma(H(t))$ remains separated from the rest of the spectrum for all t . The continuous functions $f_-(t)$ and $f_+(t)$ enclose $\sigma_*(t)$ and indicate a non-zero separation from $\sigma(H(t)) \setminus \sigma_*(t)$. Adapted from [1, p. 421]

2 RESEARCH QUESTION

How can the adiabatic theorem be effectively applied in quantum algorithms to ensure high-fidelity state transfer while maintaining computational efficiency, specifically using a modified Landau-Zener model? This investigation evaluates the practical reliability of adiabatic quantum computing by analyzing,

- **Error Quantification:** To what extent can the sharpness of analytical error bounds be used to quantify their overestimation of the exact continuous and adiabatic error?
- **Parameter Dynamics:** How do the total evolution time and the energy level spacing associated with the eigenstates of the quantum system impact the fidelity of the transition in both the continuous-time and discrete-time Schrödinger evolution?
- **Spectral Geometry Sensitivity:** How does the Hamiltonian parameter controlling the slope of the highest instantaneous eigenvalue influence the observed adiabatic error and how is this dependence reflected within the corresponding analytical error bound? Is the same sensitivity preserved in the discrete adiabatic theorem?

- **Evolution Framework:** How does the transition from a continuous-time Schrödinger evolution to a discretized approach defined by a sequence of unitary walk operators influence the accumulation of adiabatic error?

3 OBJECTIVES

The main objective of this project is to establish a structured understanding of adiabatic evolution through the following goals,

- To implement the time-depedent Hamiltonian $H(s) = [1 - f(s)]H_0 + f(s)H_1$, where $H(0)$ is the initial Hamiltonian and $H(1)$ is the target Hamiltonian. The function $f(s)$ is the scheduling function, chosen here simply as $f(s) = s$ (i.e. $f(0) = 0$ and $f(1) = 1$) to provide a smooth, fixed interpolation.
- To introduce the dimensionless time parameter $s = t/T$, which maps the evolution interval $t \in [0, T]$ to $s \in [0, 1]$ [2]
- Setup controllable parameters inside the Hamiltonian to observe the gap condition, smoothness or differentiability, and boundary behavior.
- Implement and compare the analytical error upper bound $A(s)$ [3, p. 5] for the continuous Schrödinger and discretized unitary approach with their exact adiabatic errors and interpret how they vary depending on the gap condition between the eigenstates of the corresponding energy spectrum throughout time t .
- Observe how the total evolution time T plays a vital role in controlling the error produced.

4 OVERALL SCOPE

This project focuses on adiabatic eigenstate preparation within finite-dimensional quantum systems, utilizing the time-dependent Schrödinger equation under a slowly varying Hamiltonian $H(s)$. By analyzing the adiabatic framework within spectral subspaces, the model identifies instantaneous eigenstates and maintains avoided crossings to ensure the system remains within the target subspace throughout its evolution. This continuous-time approach provides a low-error model of natural unitary dynamics, serving as a critical benchmark for the project's subsequent phases.

Building upon this foundation, the research explores error dynamics as a function of the total evolution time T and the parameters inside the time-dependent Hamiltonian, connecting these principles to broader applications such as adiabatic quantum computing, quantum simulation, and quantum annealing.

References

- [1] Stefan Teufel. *Quantum Adiabatic Theorem*, pages 419–431. Springer US, New York, NY, 2022. ISBN 978-1-0716-2621-4. doi: 10.1007/978-1-0716-2621-4_766. URL https://doi.org/10.1007/978-1-0716-2621-4_766.
- [2] Dong An, Pedro C. S. Costa, and Dominic W. Berry. Large time-step discretisation of adiabatic quantum dynamics, 2025. URL <https://arxiv.org/abs/2509.00171>.
- [3] Sabine Jansen, Mary-Beth Ruskai, and Ruedi Seiler. Bounds for the adiabatic approximation with applications to quantum computation. *Journal of Mathematical Physics*, 48(10), October 2007. ISSN 1089-7658. doi: 10.1063/1.2798382. URL <http://dx.doi.org/10.1063/1.2798382>.